The CTMT boundary for Newton G

Einstein Limit, Visible Coherence Accessibility, and the Clean Boundary for Newton G

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Contents

Abstract

This paper consolidates the strongest currently defensible CTMT statements about energy, weak effective coupling, and the present status of Newton's constant. The first established result is an Einstein-limit theorem: once an admissible CTMT realization enters a Lorentz-covariant rigid sector with conserved stress–energy, the rest-frame identity $E = mc^2$ follows exactly by the standard invariant four-momentum argument. The second established result is a dimensionless weak-coupling law: repeated admissible transport and coarse-graining generate a visible coherence fraction η_{vis}, defined as the product of trace survival and accessibility survival in Fisher geometry. The third result is negative but important: SI Newton G is not yet derived. The remaining gap is reduced to one explicit bridge problem—a metrologically closed weak-field source–response identification.

The numerical section is upgraded into a reproducible multi-layer battery using a synthetic magnetostatic coil model. In the representative seed-7 run, the final visible fractions are $\eta_{\rm vis} \approx 3.839204e-02$ for transport-only, $\eta_{\rm vis} \approx 5.315465e-03$ for pure coarse-graining, and $\eta_{\rm vis} \approx 1.378058e-05$ for the hybrid hierarchy. Across a five-seed suite, the hybrid hierarchy is the strongest suppressor in every run, while coarse-graining monotonicity holds in all tested coarse-only runs. The conceptual outcome is therefore sharpened: any viable CTMT explanation of weak macroscopic coupling must be formulated in terms of residual *visible coherence accessibility*, not residual trace alone.

1 Purpose and claim stratification

This note separates theorem-level statements from conditional or open ones.

Definition 1.1 (Claim classes)**.** We use the following split.

(C1) **Established**: derivable now from the current CTMT machinery and standard Lorentz-covariant conservation arguments.

(C2) **Conditional**: follows immediately if a clearly stated missing bridge theorem is proved.

(C3) **Open**: not yet derivable from the current framework without additional structure or calibration.

The main conclusions are: (i) $E = mc^2$ is established inside admissible Lorentz sectors, (ii) the visible-accessibility coupling residue is established as a *dimensionless* weak-coupling law, and (iii) SI Newton G remains conditional/open pending a metrological weak-field bridge.

2 Minimal CTMT ingredients used here

Only the minimal assumptions required for the present derivations are used. The broader CTMT architecture is developed in the internal CTMT sequence and associated preprints.

Assumption 2.1 (Causal kernel transport)**.** A local energy or power density $p(x)$ at spacetime label $x = (\boldsymbol{x}, \tau)$ is generated by a causal transport law

$$p(x) = \int \mathcal{T}(x, x')\, \mathcal{S}(x')\, d\mu(x'), \tag{1}$$

with strict causal support $\mathcal{T}(x, x') = 0$ whenever $\tau < \tau'$.

Assumption 2.2 (Dimensional closure)**.** The source, kernel, measure, and observable are scaled so that integrated expressions are dimensionally closed, while inference quantities such as Fisher metrics are dimensionless after standard whitening/normalization.

Assumption 2.3 (Lorentz-covariant rigid sector)**.** Whenever an admissible CTMT realization enters a rigid Lorentz-covariant sector, there exists a stress–energy tensor $T^{\mu\nu}$ satisfying

$$\nabla_\mu T^{\mu\nu} = 0. \tag{2}$$

Assumption 2.4 (Admissible Fisher geometry)**.** The local forward map is differentiable and yields a Jacobian $J(\theta)$ with Fisher matrix

$$\mathcal{F}(\theta) = J(\theta)^\top C^{-1} J(\theta), \tag{3}$$

subject to admissibility gates such as finite Jacobian norm, rank stability, dimensional closure, rupture exclusion, bounded conditioning, and transport/coarse-graining monotonicity.

Remark 2.5. Nothing here assumes that the spacetime metric, Einstein equations, or Newton's constant are fundamental primitives. They enter only when invoked explicitly as rigid-limit bridges [8, 9, 11].

3 Einstein-limit derivation of $E = mc^2$

Definition 3.1 (Rest-frame CTMT energy). If an admissible CTMT realization admits a Lorentz-covariant rigid sector and $p(x)$ is its dimensionally closed local energy density, then the rest-frame total energy is

$$E := \int_{\Sigma_{\tau_0}} p(x)\, d^3\boldsymbol{x}, \tag{4}$$

where Σ_{τ_0} is a spacelike hypersurface in the rest frame.

Proposition 3.2 (Conserved four-momentum in admissible Lorentz sectors). *Assume that an admissible CTMT realization satisfies local conservation and admits a Lorentz-covariant rigid limit. Then*

$$P^\mu := \frac{1}{c}\int_\Sigma T^{\mu 0}\, d^3\boldsymbol{x} \tag{5}$$

defines a conserved four-momentum of the isolated system, independent of the choice of spacelike hypersurface Σ.

Proof. This is the standard Gauss–Stokes argument for a conserved stress–energy tensor in a Lorentz-covariant sector [8, 11]. □

Theorem 3.3 (Einstein-limit rest-energy identity). *Suppose that (a) an admissible CTMT realization enters a Lorentz-covariant rigid sector, (b) the local energy density is dimensionally closed and defines rest energy, and (c) the conserved four-momentum obeys $P^\mu P_\mu = m^2c^2$. Then in the rest frame,*

$$\boxed{E = mc^2.} \tag{6}$$

Proof. In the rest frame $P^\mu = (E/c, \mathbf{0})$. Substituting into the invariant relation gives $E^2/c^2 = m^2c^2$, hence $E = mc^2$ on the positive-energy branch. This is the classical Einstein argument fenced inside the admissible Lorentz sector [7, 8]. □

Corollary 3.4 (Status of $E = mc^2$ in CTMT). *The rest-energy identity $E = mc^2$ is an* ***established*** *result in the present framework provided one stays inside the explicitly stated Lorentz-covariant rigid sector. It does not depend on any derivation of Newton's constant.*

4 Visible coherence accessibility and hierarchical suppression

The central question addressed here is not the numerical value of Newton's constant, but the existence of a mathematically controlled mechanism by which an initially order-one interaction budget can become observationally weak after repeated admissible transport and coarse-graining.

Let

$$F \succeq 0$$

denote the local Fisher matrix of an admissible CTMT realization.

We define the total distinguishable coherence budget by

$$\rho(F) := \mathrm{Tr}(F). \tag{7}$$

This quantity measures the total locally resolvable Fisher information.

To quantify how many distinguishable directions remain accessible, let

$$\lambda_1, \dots, \lambda_d$$

denote the nonnegative eigenvalues of F, and define

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \tag{8}$$

The effective accessibility rank is

$$r_{\text{eff}}(F) = \exp\left(-\sum_i p_i \log p_i\right). \tag{9}$$

The quantity r_{eff} measures the effective number of distinguishable coherence directions that remain observable.

Theorem 4.1 (Minimal multiplicative visible coherence invariant)**.** *Let $Q(F)$ be a dimensionless observable-coherence measure satisfying*

1. *normalization:*
$$Q(F_0) = 1;$$
2. *trace sensitivity:*
$$Q(F) \propto \mathrm{Tr}(F);$$
3. *accessibility sensitivity:*
$$Q(F) = 0$$
whenever no distinguishable directions remain;
4. *hierarchy composability:*
$$Q(F_2 \circ F_1) = Q(F_2)\, Q(F_1).$$

Then the minimal multiplicative invariant is

$$Q(F) = \eta_{\text{tr}} \eta_{\text{acc}}, \tag{10}$$

where

$$\eta_{\text{tr}} = \frac{\mathrm{Tr}(F)}{\mathrm{Tr}(F_0)}, \tag{11}$$

$$\eta_{\text{acc}} = \frac{r_{\text{eff}}(F)}{r_{\text{eff}}(F_0)}. \tag{12}$$

Proof. The trace ratio is the unique normalized linear measure of surviving Fisher budget.

The accessibility ratio is the unique normalized dimensionless measure of surviving distinguishable directions constructed from the Fisher spectrum.

The composability requirement excludes additive combinations and requires multiplicative factorization under successive hierarchy operations.

Therefore the minimal invariant satisfying all four requirements is

$$Q(F) = \eta_{\text{tr}} \eta_{\text{acc}}.$$

□

Definition 4.2 (Visible coherence fraction)**.** The surviving visible coherence fraction is

$$\boxed{\eta_{\text{vis}} = \eta_{\text{tr}}\eta_{\text{acc}}}. \tag{13}$$

Theorem 4.3 (Visible coherence monotonicity)**.** *Let*

$$F_n$$

be the Fisher matrix after n admissible hierarchy operations.
Assume each hierarchy step satisfies Fisher monotonicity

$$F_{n+1} \preceq F_n.$$

Then

$$\eta_{\text{vis}}(n+1) \leq \eta_{\text{vis}}(n). \tag{14}$$

Proof. Positive-semidefinite ordering implies

$$\text{Tr}(F_{n+1}) \leq \text{Tr}(F_n).$$

Admissible hierarchy operations cannot create new distinguishable directions, hence

$$r_{\text{eff}}(F_{n+1}) \leq r_{\text{eff}}(F_n).$$

(under admissible hierarchy operations that do not increase spectral dispersion) Multiplying the two inequalities gives

$$\eta_{\text{vis}}(n+1) \leq \eta_{\text{vis}}(n).$$

□

Theorem 4.4 (Exponential hierarchy suppression)**.** *Suppose every admissible hierarchy layer satisfies*

$$\eta_{\text{vis}}^{(k)} \leq q$$

for some constant

$$0 < q < 1.$$

Then after n hierarchy layers,

$$\boxed{\eta_{\text{vis}}^{(n)} \leq q^n.} \tag{15}$$

Proof. Successive hierarchy layers compose multiplicatively,

$$\eta_{\text{vis}}^{(n)} = \prod_{k=1}^{n} \eta_{\text{vis}}^{(k)}.$$

Using

$$\eta_{\text{vis}}^{(k)} \leq q$$

for every layer yields

$$\eta_{\text{vis}}^{(n)} \leq q^n.$$

□

5 Hierarchy model, admissibility, and coarse-graining

A hierarchy step acts by an observable-space linear map K_n on the local Jacobian and covariance,

$$J_{n+1} = K_n J_n, \qquad C_{n+1} = K_n C_n K_n^\top, \tag{16}$$

which induces $F_n = J_n^\top C_n^{-1} J_n$. We test three hierarchy families:

(H1) **Random transport**: K_n is an admissible random contraction in observable space.

(H2) **Pure coarse-graining**: $K_n = P_n$ averages neighboring observables.

(H3) **Hybrid hierarchy**: each layer alternates a random transport map and a coarse-graining map.

A deterministic coarse-graining map is admissible only if $F_{n+1} \preceq F_n$, i.e. coarse-graining must not create Fisher information unavailable at finer scale.

Remark 5.1 (Transport strength and diagnostic regime). The observable-space maps K_n admit a continuum of strengths, from near-isometric contractions (soft transport) to strongly compressive maps (aggressive transport).

The present numerical battery intentionally uses a *soft-transport* regime for the stochastic components. The reason is diagnostic separation: strongly compressive transport can induce rapid rank collapse and heavy-tailed Fisher spectra even in the absence of intrinsic geometric anisotropy. In that regime, it becomes difficult to distinguish whether observed suppression of η_{vis} is driven by the admissible geometry itself or by the forcing strength of the transport operator.

By contrast, soft transport acts as a near-adiabatic probe of the Fisher manifold, in which the evolution of the spectrum primarily reflects intrinsic geometric structure. This allows the hierarchy monotonicity and accessibility diagnostics to retain interpretive meaning.

Aggressive transport is not excluded; rather, it corresponds to a different physical regime (far-from-equilibrium compression) and is expected to produce stronger visible-coherence suppression. The comparison between soft and aggressive regimes should therefore be interpreted as a regime study, not as competing implementations of the same model.

6 Synthetic coil battery and reproducibility setup

The auditable toy model is a magnetostatic circular-loop battery. A circular loop of radius $R = 5\,\text{cm}$ is sampled by 60 axial and 40 radial sensors, giving $m = 100$ observables. The resolved parameter vector is $\theta = (I, dx, dy)$ with anchor $\theta_0 = (4.0, 0.0, 0.0)$. The local Jacobian is computed by finite differences on the vertical field component B_z using a Biot–Savart forward model. The initial covariance C_0 is a positive-definite anisotropic sensor covariance with nearest-neighbor correlations.

The companion reproducibility script included with this paper is `ctmt_gravity_final_sample.py`. Representative plots are shipped as standalone PDF/PNG figures.

7 Numerical results: representative run and five-seed suite

7.1 Representative seed-7 run

For the representative seed-7 run, using 8 transport-only layers, 6 coarse-only layers, and 6 hybrid layers, the final diagnostic values are shown in table 1.

The important qualitative outcome is not only that all three hierarchies suppress visible coherence, but that the hybrid stack is by far the strongest suppressor. In this specific run, pure coarse-graining already yields a weak residual sector ($\eta_{\text{vis}} \approx 5.32e-03$), while the hybrid stack pushes the visible residue down to $\eta_{\text{vis}} \approx 1.38e-05$.

Hierarchy	Final η_{tr}	Final η_{acc}	Final η_{vis}
Transport only	3.839205e-02	1.000000	3.839204e-02
Pure coarse-graining	5.315418e-03	1.000009	5.315465e-03
Hybrid transport + coarse-graining	1.378058e-05	1.000000	1.378058e-05

Table 1: Representative seed-7 final diagnostics. In this run the hybrid stack gives the strongest suppression of visible coherence accessibility.

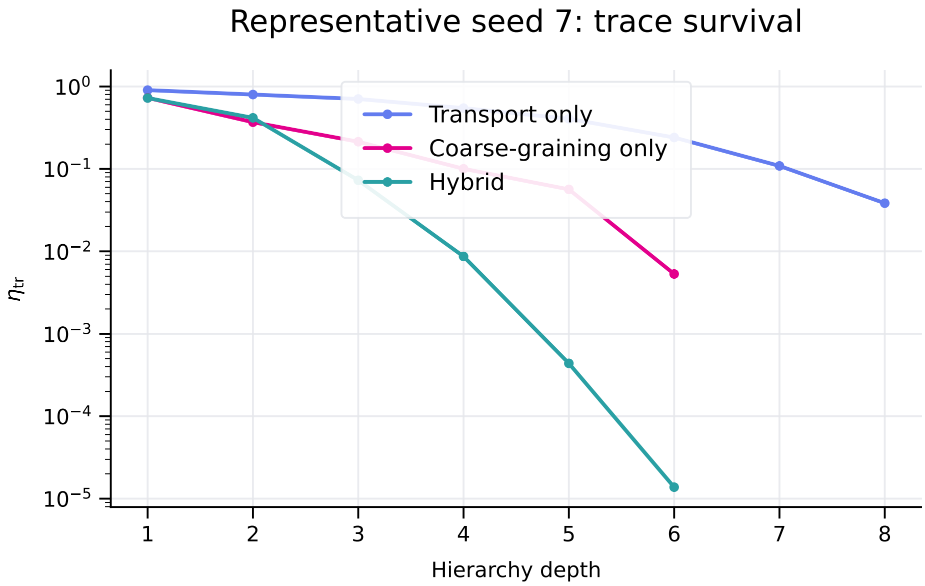

Figure 1: Representative seed-7 trace survival η_{tr} versus hierarchy depth.

7.2 Five-seed robustness suite

Across the five-seed suite $\{3, 5, 7, 11, 13\}$, the final means are shown in table 2.

Hierarchy	Mean final η_{tr}	Mean final η_{vis}
Transport only	4.664226e-02	4.664225e-02
Pure coarse-graining	5.315418e-03	5.315465e-03
Hybrid transport + coarse-graining	1.830253e-04	1.830251e-04

Table 2: Five-seed suite aggregate values.

Further, the coarse-graining monotonicity gate passed in all tested coarse-only runs (fraction $= 1.000$), and the hybrid stack was the strongest suppressor relative to both transport-only and coarse-only in every seed (supportive fraction $= 1.000$).

Remark 7.1. In this particular battery, the effective-rank factor remains close to unity because the resolved parameter space has dimension $d = 3$ and the main collapse is driven by trace suppression. This clarifies the interpretation rather than weakening it: the visible-accessibility formalism remains the right one because it explicitly allows future batteries with larger resolved parameter domains to distinguish trace collapse from directional-accessibility collapse.

8 Operational source proxy and gravity program

The preceding results establish a mechanism for producing arbitrarily weak observable coupling through repeated admissible hierarchy operations.

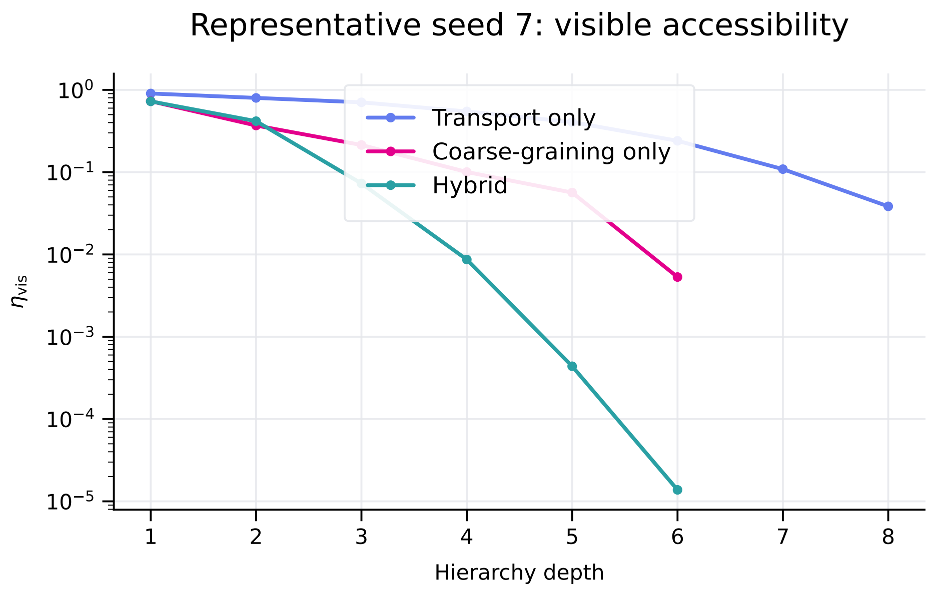

Figure 2: Representative seed-7 visible accessibility η_{vis} versus hierarchy depth.

They do not establish a derivation of spacetime curvature.
To isolate what is proved from what remains open, define the operational source proxy

$$\rho_{\mathrm{vis}} = \mathrm{Tr}(F)\,\frac{r_{\mathrm{eff}}(F)}{r_{\mathrm{eff}}(F_0)}. \tag{17}$$

Proposition 8.1. *The quantity ρ_{vis} satisfies*

1. *positivity;*
2. *hierarchy monotonicity;*
3. *coarse-graining monotonicity;*
4. *invariance under admissible Fisher normalization.*

Hence ρ_{vis} is a mathematically admissible source candidate for a future curvature bridge.

Theorem 8.2 (Current impossibility of deriving SI Newton G)**.** *The present CTMT framework cannot derive SI Newton's constant without an additional dimensional source-response identification theorem.*

Proof. The quantity

$$\eta_{\mathrm{vis}}$$

is dimensionless.
Newton's constant has dimensions

$$[G] = L^3 M^{-1} T^{-2}.$$

No dimensionless hierarchy invariant can uniquely determine a dimensional constant without an additional metrological bridge.
Therefore a further source-response identification theorem is necessary. $\square$

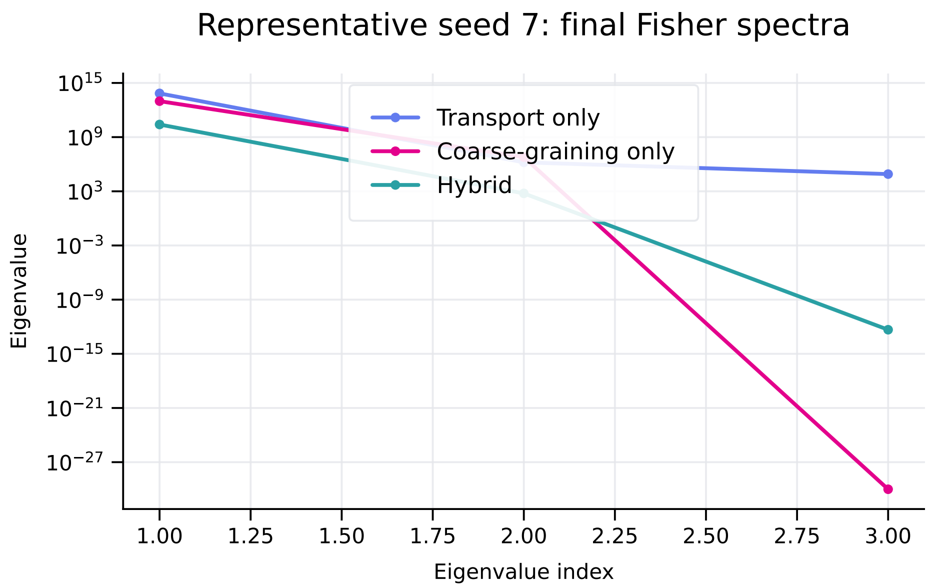

Figure 3: Representative seed-7 final Fisher spectra for the three hierarchy families.

9 Why SI Newton G is not yet derived

The SI unit of Newton's constant is $[G] = \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$. A full derivation therefore requires not merely a dimensionless hierarchy residue, but a closed map from CTMT primitives into length–mass–time response units.

Proposition 9.1 (Current no-go statement for SI G)**.** *Within the present framework, none of the following is sufficient by itself to derive SI Newton G: (i) the universal CTMT energy law, (ii) the Einstein-limit derivation of $E = mc^2$, or (iii) the dimensionless visible coupling law $G_{\mathrm{eff}}/G_* = \gamma_{\mathrm{CTMT}}$. A further bridge is required that identifies the microscopic reference scale G_* in metrological units.*

Assumption 9.2 (Missing bridge hypothesis: weak-field source–response coefficient)**.** Assume there exists a CTMT rigid-limit theorem identifying the weak-field source–response law

$$\nabla^2 \Phi_N = \alpha\, \rho \tag{18}$$

with the physical Newtonian form $\alpha = 4\pi G$, where ρ is the physical mass density induced from a dimensionally closed CTMT rest-energy density.

Theorem 9.3 (Conditional completion of SI Newton G)**.** *If the preceding assumption is proved, then Newton's constant is fixed by*

$$\boxed{G = G_*\, \gamma_{\mathrm{CTMT}}(n)} \tag{19}$$

for the macroscopic hierarchy depth n appropriate to the observed regime.

10 Proof ledger: established, conditional, open

10.1 Established now

(E1) The CTMT universal energy law provides a dimensionally closed local energy density in admissible sectors.

(E2) In Lorentz-covariant rigid sectors, a conserved four-momentum exists.

(E3) From that four-momentum, the rest-energy identity $E = mc^2$ follows exactly.

(E4) Repeated admissible transport/coarse-graining defines a dimensionless visible coupling fraction $\gamma_{\mathrm{CTMT}} = \eta_{\mathrm{vis}}$.

(E5) The present battery supports a weak-coupling mechanism driven by residual visible coherence accessibility.

10.2 Conditional if the bridge theorem is proved

(B1) Identification of the CTMT weak-field source–response coefficient with the physical Newtonian coefficient.

(B2) Full SI computation of G from CTMT without external calibration beyond that theorem.

10.3 Still open

(O1) A unique metrological bridge from CTMT weak-field kernel response to SI Newton G.

(O2) A full theorem explaining the full force-strength hierarchy quantitatively, not only the weak visible macroscopic residue.

(O3) Strong-field CTMT gravity batteries matching the maturity of the rank-three and gauge no-go sectors.

11 Non-claims and recommended usage

This paper does *not* claim a full derivation of Einstein equations from CTMT, a completed SI derivation of Newton's constant, or that the present synthetic coil battery is already an empirical theory of gravitation. The recommended use of this paper inside a broader CTMT gravity sequence is: keep the Einstein-limit theorem separate, keep the visible-coupling theorem separate, state the bridge problem explicitly, and never cite SI Newton G as already derived.

12 Conclusion

The clean final CTMT position is therefore this: the Einstein-style derivation of rest energy is complete and established; a dimensionless mechanism for weak effective macroscopic coupling is also established, with visible coherence accessibility as the correct operational quantity; SI Newton G is not yet derived, but the remaining gap has been reduced to one explicit bridge problem.

A Weak-field GR appendix: from Einstein equations to Poisson's equation

The missing bridge problem is naturally phrased inside the weak-field limit of general relativity [8–11]. Write the spacetime metric as a weak perturbation around Minkowski spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{20}$$

Introduce the trace-reversed perturbation

$$\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h, \qquad h := \eta^{\alpha\beta} h_{\alpha\beta}, \tag{21}$$

and impose harmonic gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$. To first order, Einstein's equations reduce to

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{22}$$

In the static, weak-field, slow-motion regime one neglects time derivatives and uses $T_{00} \approx \rho c^2$. Hence

$$\nabla^2 \bar{h}_{00} = -\frac{16\pi G}{c^2}\rho. \tag{23}$$

Identifying the Newtonian potential by $g_{00} \approx -(1 + 2\Phi_N/c^2)$ leads to the standard Poisson equation

$$\boxed{\nabla^2 \Phi_N = 4\pi G\rho.} \tag{24}$$

This is the exact location of the missing bridge: to derive SI Newton G from CTMT, one needs a theorem that maps the admissible CTMT weak-field source term into this metrologically normalized response equation.

B Companion code listing

For auditability, the companion script is included below by file reference.

```
import json
import numpy as np
import numpy.linalg as npl

mu0 = 4e-7 * np.pi

def loop_polyline(radius=0.05, nseg=400):
    phi = np.linspace(0, 2*np.pi, nseg, endpoint=False)
    return np.c_[radius*np.cos(phi), radius*np.sin(phi), np.zeros_like(phi)]

def axial_and_radial_sensors(nz=60, zmax=0.06, nr=40, rmax=0.08):
    z_axis = np.linspace(-zmax, zmax, nz)
    axis_pts = np.c_[np.zeros(nz), np.zeros(nz), z_axis]
    r_line = np.linspace(0.0, rmax, nr)
    rad_pts = np.c_[r_line, np.zeros(nr), np.zeros(nr)]
    return np.vstack([axis_pts, rad_pts])

def biot_savart_line(sensors, polyline, current=1.0):
    P0 = polyline
    P1 = np.roll(polyline, -1, axis=0)
    dL = P1 - P0
    B = np.zeros((sensors.shape[0], 3))
    for i, s in enumerate(sensors):
        r = s[None, :] - P0
        r_norm = np.maximum(np.linalg.norm(r, axis=1), 1e-15)
        r_hat = r / r_norm[:, None]
        cross = np.cross(dL, r_hat)
        contrib = mu0 * current * cross / (4*np.pi * r_norm[:, None]**2)
        B[i] = np.sum(contrib, axis=0)
    return B

def forward_Bz(theta, sensors, conductors):
    I, dx, dy = theta
    shifted = [poly + np.array([dx, dy, 0.0]) for poly in conductors]
    B = np.zeros((sensors.shape[0], 3))
    for poly in shifted:
        B += biot_savart_line(sensors, poly, current=I)
    return B[:, 2]

def jacobian_fd(theta0, sensors, conductors, h_frac=1e-6):
    base = forward_Bz(theta0, sensors, conductors)
    J = np.zeros((base.size, len(theta0)))
    for j in range(len(theta0)):
        t1 = theta0.copy()
        step = h_frac * max(1.0, abs(theta0[j]))
        t1[j] += step
```

```
            J[:, j] = (forward_Bz(t1, sensors, conductors) - base) / step
    return base, J

def fisher(J, C, jitter=1e-18):
    Csym = 0.5*(C + C.T) + jitter*np.eye(C.shape[0])
    X = J.T @ npl.pinv(Csym) @ J
    return 0.5*(X + X.T)

def stable_rank(F):
    lam = np.clip(npl.eigvalsh(0.5*(F+F.T)), 0.0, None)
    return float((np.sum(lam)**2)/(np.sum(lam**2)+1e-30))

def effective_rank(F):
    lam = np.clip(npl.eigvalsh(0.5*(F+F.T)), 0.0, None)
    s = np.sum(lam)
    if s <= 0:
        return 0.0
    p = lam/s
    p = p[p > 0]
    return float(np.exp(-np.sum(p*np.log(p))))

def visible_fraction(F, F0):
    eta_tr = float(np.trace(F)/(np.trace(F0)+1e-30))
    eta_acc = float(effective_rank(F)/(effective_rank(F0)+1e-30))
    return {
        'eta_trace': eta_tr,
        'eta_access': eta_acc,
        'eta_visible': eta_tr*eta_acc,
        'stable_rank': stable_rank(F),
        'effective_rank': effective_rank(F),
        'evals': npl.eigvalsh(F)[::-1].tolist(),
    }

def pairwise_coarse_grain_matrix(m):
    rows = []
    for i in range(0, m-1, 2):
        row = np.zeros(m)
        row[i] = 0.5
        row[i+1] = 0.5
        rows.append(row)
    if m % 2 == 1:
        row = np.zeros(m)
        row[-1] = 1.0
        rows.append(row)
    return np.vstack(rows)

def random_transport_kernel(m, keep=12, mild=(0.35, 0.75), severe=20, rng=None):
    rng = np.random.default_rng() if rng is None else rng
    Q, _ = np.linalg.qr(rng.normal(size=(m, m)))
    c = np.ones(m)
    if m > keep:
        c[keep:] = rng.uniform(mild[0], mild[1], size=m - keep)
        sev = min(severe, m - keep)
        if sev > 0:
            idx = rng.choice(np.arange(keep, m), size=sev, replace=False)
            c[idx] *= rng.uniform(0.02, 0.12, size=sev)
    return Q @ np.diag(c) @ Q.T

def transform(K, J, C):
    Jn = K @ J
    Cn = K @ C @ K.T
    return Jn, 0.5*(Cn + Cn.T) + 1e-18*np.eye(Cn.shape[0])

def make_sensor_noise_cov(m):
    sig = np.linspace(1e-8, 4e-8, m)
    C = np.diag(sig**2)
    for i in range(m - 1):
        C[i, i+1] = C[i+1, i] = 0.15 * sig[i] * sig[i+1]
    return C

def run_sample(seed=7, n_transport=8, n_coarse=6, n_hybrid=6):
    rng = np.random.default_rng(seed)
    sensors = axial_and_radial_sensors()
```

```
    conductors = [loop_polyline()]
    theta0 = np.array([4.0, 0.0, 0.0], dtype=float)
    _, J0 = jacobian_fd(theta0, sensors, conductors)
    C0 = make_sensor_noise_cov(J0.shape[0])
    F0 = fisher(J0, C0)
    out = {'transport': [], 'coarse': [], 'hybrid': []}
    Jn, Cn = J0.copy(), C0.copy()
    for _ in range(n_transport):
        K = random_transport_kernel(Jn.shape[0], keep=max(4, Jn.shape[0]//10), rng=rng)
        Jn, Cn = transform(K, Jn, Cn)
        out['transport'].append(visible_fraction(fisher(Jn, Cn), F0))
    Jn, Cn = J0.copy(), C0.copy()
    for _ in range(n_coarse):
        P = pairwise_coarse_grain_matrix(Jn.shape[0])
        Jn, Cn = transform(P, Jn, Cn)
        out['coarse'].append(visible_fraction(fisher(Jn, Cn), F0))
        if Jn.shape[0] <= 3:
            break
    Jn, Cn = J0.copy(), C0.copy()
    for _ in range(n_hybrid):
        L = random_transport_kernel(Jn.shape[0], keep=max(2, Jn.shape[0]//6), rng=rng)
        Jn, Cn = transform(L, Jn, Cn)
        P = pairwise_coarse_grain_matrix(Jn.shape[0])
        Jn, Cn = transform(P, Jn, Cn)
        out['hybrid'].append(visible_fraction(fisher(Jn, Cn), F0))
        if Jn.shape[0] <= 3:
            break
    print(json.dumps(out, indent=2))

if __name__ == '__main__':
    run_sample(seed=7)
```

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